GUIDELINES FOR VALIDATION OF ELECTRIC POWER TABLES FOR EEDI (EPT-EEDI)

1 INTRODUCTION

The purpose of these Guidelines is to assist Recognized Organizations in the validation of ship's Electric Power Tables (EPT) for calculation of Energy Efficiency Design Index (EEDI). As such, these Guidelines support the implementation of "GUIDELINES ON THE METHOD OF CALCULATION OF THE ATTAINED ENERGY EFFICIENCY DESIGN INDEX FOR NEW SHIPS" ("EEDI Calculation Guidelines ", hereafter) and "GUIDELINES ON SURVEY AND

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2 OBJECTIV ES

These Guidelines provide a framework for the uniform application of the EPT-EEDI validation process for those ships that come under paragraph 2.5.6.3 of the EEDI Calculation Guidelines.

3 DEFINITIONS

3.1 Applicant means an organization, primarily a shipbuilder or a ship designer, which requests the EPT-EEDI validation in accordance with these Guidelines.

3.2 Validator means a Recognized Organization which conducts the EPT-EEDI validation in accordance with these Guidelines.

3.3 Validation for the purpose of these Guidelines means review of submitted documents and survey during construction and sea trials.

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3.4 Standard EPT- EEDI-Form refers to the layout given in appendix 3 that contains the EPT-EEDI results that will be the subject of validation. Other supporting documents submitted for this purpose will be used as reference only and will not be subject to validation.

3.5 P AE herein is defined as per definition in paragraph 2.5.6 of the EEDI Calculation Guidelines.

3.6 Ship Service and Engine-room Loads refer to all the load groups which are needed for the hull, deck, navigation and safety services, propulsion and auxiliary engine services, engine-room ventilation and auxiliaries and ship's general services.

3.7 Diversity Factor is the ratio of the "total installed load power" and the "actual load power" for continuous loads and intermittent loads. This factor is equivalent to the product of service factors for load, duty and time.

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4 APPLICATION

4.1 These Guidelines are applicable to ships as stipulated by paragraph 2.5.6.3 of the EEDI Calculation Guidelines.

4.2 These Guidelines should be applied for new ships for which an application for an EPT-EEDI validation has been submitted to a validator.

4.3 The steps of the validation process include:

.1 Review of documents during the design stage

.1.1 Check if all relevant loads are listed in the EPT;
.1.2 Check if reasonable service factors are used; and
.1.3 Check the correctness of the P AE calculation based on the data given in the EPT.

.2 Survey of installed systems and components during construction stage

.2.1 Check if a randomly selected set of installed systems and components are correctly listed with their characteristics in the EPT.

.3 Survey of sea trials

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.2 Survey of installed systems and components during construction stage

.2.1 Check if a randomly selected set of installed systems and components are correctly listed with their characteristics in the EPT.

.3 Survey of sea trials

.3.1 Check if selected units/loads specified in EPT are observed.

5 SUPPORTING DOCUMENTS

5.1 The applicant should provide as a minimum the ship electric balance load analysis.

5.2 Such information may contain shipbuilders' confidential information. Therefore, after the validation, the validator should return all or part of such information to the applicant at the applicant's request.

5.3 A special EEDI condition during sea trials may be needed and defined for each vessel and included into the sea trial schedule. For this condition a special column should be inserted into the EPT.

6 PROCEDURES FOR VALIDATION

6.1 General

6.1.1 P AE should be calculated in accordance with the EPT-EEDI Calculation Guidelines. EPT-EEDI validation should be conducted in two stages: preliminary validation at the design stage and final validation during sea trials. The validation process is presented in figure 1.

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Figure 1: Basic Flow of EPT-EEDI Validation Process

6.2 Preliminary validation at the design stage

6.2.1 For the preliminary validation at the design stage, the applicant should submit to a validator an application for the validation of EPT-EEDI inclusive of EPT-EEDI Form and all the relevant and necessary information for the validation as supporting documents.

6.2.2 The applicant should supply as a minimum the supporting data and information, as specified in appendix A (to be developed).

6.2.3 The validator may request from the applicant additional information on top of those contained in these Guidelines, as necessary, to enable the validator to examine the calculation process of the EPT-EEDI. The estimation of the ship EPT-EEDI at the design

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contained in these Guidelines, as necessary, to enable the validator to examine the calculation process of the EPT-EEDI. The estimation of the ship EPT-EEDI at the design stage depends on each applicant's experiences, and it may not be practicable to fully examine the technical aspects and details of each machinery component. Therefore, the preliminary validation should focus on the calculation process of the EPT-EEDI that should follow the best marine practices.

Note: A possible way forward for more robust validation is to establish a standard methodology of deriving the ship EPT by setting standard formats as agreed and used by industry.

Applicant (Primarily shipbuilder or ship designer) Va lidator (Recognised Organization such as Class Society)
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